3.1 Classwork: The Unit Circle and Trigonometric Graphs (no calculator) — Markscheme

1. [Maximum mark: 14]

(a) Angles and coordinates $(\cos\theta,\ \sin\theta)$:

0: $(1,\ 0)$	$\frac{\pi}{2}$: $(0,\ 1)$	π: $(-1,\ 0)$	$\frac{3\pi}{2}$: $(0,\ -1)$
$\frac{\pi}{6}$: $\left(\frac{\sqrt{3}}{2},\ \frac{1}{2}\right)$	$\frac{2\pi}{3}$: $\left(-\frac{1}{2},\ \frac{\sqrt{3}}{2}\right)$	$\frac{7\pi}{6}$: $\left(-\frac{\sqrt{3}}{2},\ -\frac{1}{2}\right)$	$\frac{5\pi}{3}$: $\left(\frac{1}{2},\ -\frac{\sqrt{3}}{2}\right)$
$\frac{\pi}{4}$: $\left(\frac{\sqrt{2}}{2},\ \frac{\sqrt{2}}{2}\right)$	$\frac{3\pi}{4}$: $\left(-\frac{\sqrt{2}}{2},\ \frac{\sqrt{2}}{2}\right)$	$\frac{5\pi}{4}$: $\left(-\frac{\sqrt{2}}{2},\ -\frac{\sqrt{2}}{2}\right)$	$\frac{7\pi}{4}$: $\left(\frac{\sqrt{2}}{2},\ -\frac{\sqrt{2}}{2}\right)$
$\frac{\pi}{3}$: $\left(\frac{1}{2},\ \frac{\sqrt{3}}{2}\right)$	$\frac{5\pi}{6}$: $\left(-\frac{\sqrt{3}}{2},\ \frac{1}{2}\right)$	$\frac{4\pi}{3}$: $\left(-\frac{1}{2},\ -\frac{\sqrt{3}}{2}\right)$	$\frac{11\pi}{6}$: $\left(\frac{\sqrt{3}}{2},\ -\frac{1}{2}\right)$

A1A1A1A1A1A1

A1 for all 16 angles in radians at the correct points. A1 for the four points on the axes. A1 for each quadrant's three non-axis points (first, second, third, fourth), with both coordinates and both signs correct. Accept $\frac{1}{\sqrt{2}}$ for $\frac{\sqrt{2}}{2}$. Angles in degrees earn the first A0. Swapping the coordinates of the $\frac{\pi}{6}$ and $\frac{\pi}{3}$ points is the common error: A0 for that quadrant.

(b) $\sin\frac{5\pi}{6} = \frac{1}{2}$ **A1**

$\cos\frac{4\pi}{3} = -\frac{1}{2}$ **A1**

$\tan\frac{7\pi}{4} = -1$ **A1**

Common errors, each A0: $\sin\frac{5\pi}{6} = -\frac{\sqrt{3}}{2}$ *(the cosine),* $\cos\frac{4\pi}{3} = \frac{1}{2}$ *(sign),* $\tan\frac{7\pi}{4} = 1$ *(sign).*

(c) $\sin^2\frac{5\pi}{6} + \cos^2\frac{5\pi}{6} = \left(\frac{1}{2}\right)^2 + \left(-\frac{\sqrt{3}}{2}\right)^2$ **M1**

$= \frac{1}{4} + \frac{3}{4}$ **A1**

$= 1$ **AG**

M1 for substituting both exact values. A1 for $\frac{1}{4}$ and $\frac{3}{4}$ seen. Accept $\frac{\sqrt{3}}{2}$ without the negative sign for the M1, since it is squared, but $-\frac{3}{4}$ for its square earns A0.

(d) $\cos^2\theta = 1 - \left(\frac{2}{3}\right)^2 = \frac{5}{9}$ **M1**

$\cos\theta = \pm\frac{\sqrt{5}}{3}$ **A1**

$\cos\theta = -\frac{\sqrt{5}}{3}$, since θ is in the second quadrant **A1**

M1 for using the identity with $\sin\theta = \frac{2}{3}$*. The final A1 is for choosing the negative value;* $+\frac{\sqrt{5}}{3}$ *as the final answer earns M1A1A0. Wrong methods:* $1 - \frac{2}{3} = \frac{1}{3}$ *(squaring forgotten) gives* $\cos\theta = -\frac{1}{\sqrt{3}}$*, earning M0A0A0.*

2. [Maximum mark: 11]

(a) Points $(0,\ 0)$, $\left(\frac{\pi}{6},\ 0.5\right)$, $\left(\frac{\pi}{3},\ 0.87\right)$, $\left(\frac{\pi}{2},\ 1\right)$, …, $(\pi,\ 0)$, …, $\left(\frac{3\pi}{2},\ -1\right)$, …, $(2\pi,\ 0)$ plotted **A1**

Smooth wave through the points, labelled $y = \sin x$ **A1**

(b) Points $(0,\ 1)$, $\left(\frac{\pi}{6},\ 0.87\right)$, $\left(\frac{\pi}{3},\ 0.5\right)$, $\left(\frac{\pi}{2},\ 0\right)$, …, $(\pi,\ -1)$, …, $(2\pi,\ 1)$ plotted **A1**

Smooth wave through the points, labelled $y = \cos x$ **A1**

In (a) and (b), the first A1 needs at least 11 of the 13 points within half a grid square. The second A1 needs a smooth curve (no straight segments, no corners at the maxima and minima) and a label. Curves labelled the wrong way round lose the second A1 in both (a) and (b).

(c) gradient of $OP = \dfrac{\sin\theta - 0}{\cos\theta - 0}$ **M1**

$= \dfrac{\sin\theta}{\cos\theta}$ **A1**

$= \tan\theta$ **AG**

M1 for rise over run using the coordinates of O and P. A1 for $\frac{\sin\theta}{\cos\theta}$ before the given answer. Accept a right-triangle argument: opposite $\sin\theta$, adjacent $\cos\theta$.

(d) $m = \tan\frac{\pi}{6} = \frac{1}{\sqrt{3}}$ **A1**

$y = \frac{1}{\sqrt{3}}x + 2$ **A1**

Accept $\frac{\sqrt{3}}{3}$ for the gradient. FT the equation from the candidate's gradient. Wrong methods: $\tan\frac{\pi}{6} = \sqrt{3}$ (the value for $\frac{\pi}{3}$), or $m = \sin\frac{\pi}{6} = \frac{1}{2}$, earn A0, then A1 FT for $y = mx + 2$.

(e) $m = \tan\frac{2\pi}{3}$ **M1**

$= -\sqrt{3}$ **A1**

$y - 4 = -\sqrt{3}(x - 1) \quad (\text{so } y = -\sqrt{3}x + 4 + \sqrt{3})$ **A1**

Accept any correct form of the equation. FT the equation from the candidate's gradient. Wrong methods: $m = +\sqrt{3}$ (sign) earns M1A0A1 FT; swapping the coordinates of the point, $y - 1 = -\sqrt{3}(x - 4)$, earns M1A1A0.